

RECESS FINAL PROJECT 2026  
BSE2301 SOFTWARE ENGINEERING MINI PROJECT 2

Having successfully completed the intricacies of three core areas in python programming: Data Science with Python, Machine Learning basics, and Web Development with Python. Each section is designed to challenge software engineering students. By working through these projects, you will enhance your understanding and gain hands-on experience, preparing you for real-world applications and advanced problem-solving in the tech industry.

Use python and analytics functions to interrogate the data given to you and derive actionable insights about it. You will be required to include most of the functions you learn. Using functions that were not in the lecture scope will even earn your team higher marks.

### **PART A: DATA SCIENCE AND MACHINE LEARNING**

You will be expected to submit the **Data Science and Machine Learning** projects as guided;

1. A final report of not more than 10 pages (Soft and Hard Copy Documentation) including;

An introduction about your dataset.

**Task 1:** Explore the dataset, identify missing values, and determine the percentage of missing data for each feature.

**Task 2:** Decide on the appropriate method for handling missing data (imputation, removal, or using algorithms that handle missing values natively) and justify your choice.

**Task 3:** Implement the chosen method and evaluate its impact on the dataset

**Task 4:** Explore the dataset and identify potential features.

**Task 5:** Implement feature engineering techniques to create new features.

**Task 6:** Evaluate the impact of the new features on model performance.

**Task 7:** Explore the dataset and identify key variables for visualization.

**Task 8:** Use libraries like Matplotlib, Seaborn, and Plotly to create various types of visualizations (e.g., scatter plots, heatmaps, 3D plots).

**Task 9:** Interpret the visualizations to uncover patterns and insights in the data.

**Task 10:** Data Splitting, Training of modules

**Task 11:** Cross Validation and Evaluation of the model

**Task 12:** Conclusions, Actionable insights, Recommendation and summary

The key objectives for analyzing your data.

(Also include pictures and explanations of your data before and after cleaning it.)

Well annotated screenshots of the visualizations (plots/tables) you will have generated.  
Deductions/conclusions for your data.

2. Your python script files.

### **PART 2: WEB DEVELOPMENT PROJECTS (DJANGO AND FLASK)**

-A final report of not more than 10 pages (soft and hard copy documentation) including;

- An introduction about your Web application.
- Screenshots of your Projects
- Documentation of your web application systems
- The source code of your project via Link to the GitHub

**Note:**

This is group work and every group member should be actively involved in doing it.

The project will run for 2 weeks starting on **17th July 2026 to 30th July 2026 till 11:59 pm.**

Email us : jeff.geoff.mis@gmail.com cc. ndigezzalivingstone2@gmail.com and send us the link for the GitHub Repository!

Every group will be supervised at least once. Each group will be required to setup a convenient date and use the class link shared in the Group Leader. Supervision is mainly to track your progress and guide you where you are having challenges.

Lastly, you will have a physical presentations on an agreed date before **30th of July 2026** that will be communicated to you through the class Rep.

**PROJECT ALLOCATION**

| No. | Project Category                                               | Groups Day | Group Evening |
|-----|----------------------------------------------------------------|------------|---------------|
| 1   | Health resource allocation (DS) or Deforestation tracking (DS) | A,I, M     | A             |
| 2   | Job Recommendation Engine (Django)                             | B, X, P, H | J             |
| 3   | AI-Powered Web Service (Flask)                                 | C, O       | B             |
| 4   | Finance credit scoring (DS)                                    | D, G       | H             |
| 5   | Real-Time Fraud Detection System (Django)                      | E,K        | F             |
| 6   | Business Intelligence Platform (Flask)                         | F, R       | R             |
| 7   | Grid load prediction (ML) or Tourism Game Park Detection (ML)  | D          | G             |
| 8   | Translator App (Django)                                        | T, E       | V             |
| 9   | Microservice Architecture (Flask)                              | L, S J     | I             |
| 10  | Language translation models (ML)                               | Q, V       |               |
| 11  | Resume Builder Application (Django)                            | W          |               |
| 12  | Forex Trading Academy Platform (Flask)                         | U, N       |               |
|     |                                                                |            |               |

**1A: DATA SCIENCE**

1. Health resource allocation
2. Deforestation tracking
3. Finance credit scoring

**1B: MACHINE LEARNING**

1. Grid load prediction
2. Tourism Game Park Detection
3. Language translation models

Download datasets from: <https://ryeko.org/datasets/>

## 2A: DJANGO PROJECTS

### Job Recommendation Engine (Django App)

You'll build a tool that suggests job openings based on a user's profile or past applications. It might use collaborative filtering or a simpler keyword-based approach. As you handle user skills, job categories, and experience levels, you'll learn to customize search results so people can find relevant listings faster.

You can also add filters like location or salary range. Over time, the system can improve at suggesting openings by tracking clicks or applications.

#### Features

- a) **Data Matching:** You will design algorithms or scoring rules to pair candidates with relevant jobs.
- b) **User Preference Storage:** You will keep track of what jobs users are interested in.
- c) **Search & Filtering:** You will let people refine results by location, pay, or job type.
- d) **Database Design:** You will manage many job postings and user profiles in a structured format

### Real-Time Fraud Detection System (Django App)

You will set up a tool that monitors financial transactions (like online payments or bank transfers) and flags anything that looks suspicious. It could be transactions over a certain limit or from unusual locations.

You will also design a dashboard to show a live feed of transactions, run checks, and issue alerts when certain conditions are met. You can experiment with storing patterns of past behavior and comparing them against new entries.

#### Features

- a) **Data Monitoring:** You'll track continuous entries in a database and check each one in real time.
- b) **Alerts & Notifications:** You'll send immediate warnings to administrators or relevant staff.
- c) **Anomaly Detection:** You'll compare new transactions against past patterns for red flags.
- d) **Dashboard Development:** You'll create a clear interface to watch stats and flagged transactions.

### Translator App (Django)

You will develop a web tool where users can input text and get it converted into another language. This involves connecting with a translation API and then parsing and returning the output to the user. It's a great introduction to handling external services and different language encodings.

Django view handles the logic of calling the translator API. Once the results come back, you display them for the user to copy or refine further. You might also let users save recent translations or even auto-detect the input language.

#### Features

- a) **API Integration:** You will link with a translation service that processes input text.
- b) **Data Encoding:** You will ensure correct encoding so certain characters in different scripts aren't lost.
- c) **Form Submissions:** You will gather the user's original text and target language.
- d) **Optional History Feature:** You can store previous translations for easy reuse.

### Resume Builder Application (Django)

A Resume Builder Application helps you quickly create a professional looking resume without struggling with complicated formatting. Instead of designing everything from the start, you just have to enter your details, and the app organizes them neatly into a resume.

**Key Features of the Project:**

1. On this website, you can fill in your personal details, education details, and work details one by one, in a step by step manner.
2. You can pick from different ready made professional resume designs.
3. You can see a preview of your resume before you download it.
4. You can download your resume as a PDF file.
5. You can create an account so your resume is saved, and you can update it anytime

**2B: FLASK PROJECTS****AI-Powered Web Service (Flask)**

Build a web interface for an AI model that processes user input and returns intelligent results. This could be anything from an image classifier to a text generator or a recommendation system.

**Business Intelligence Platform (Flask)**

Build a comprehensive data dashboard that connects to databases, processes data, and presents insights through interactive visualizations. This application should handle large datasets and provide real-time analytics.

**Microservice Architecture (Flask)**

Build a microservice that handles a specific business function (e.g., authentication, payment processing, or notification service) with full DevOps tooling, containerization, and orchestration.

**Forex Trading Academy Platform (Flask)**

You are required to build a Production-Ready Forex Trading Academy Platform using Flask. This platform will serve as a comprehensive educational and trading simulation system where users can learn forex trading, access educational resources, practice with virtual money, and track their progress.

**\*\*\*Happy Coding\*\*\***